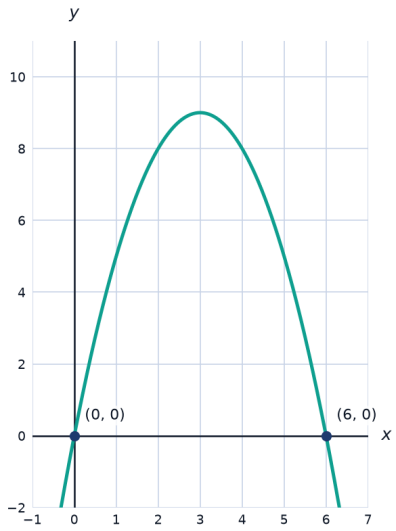
Integration and Kinematics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

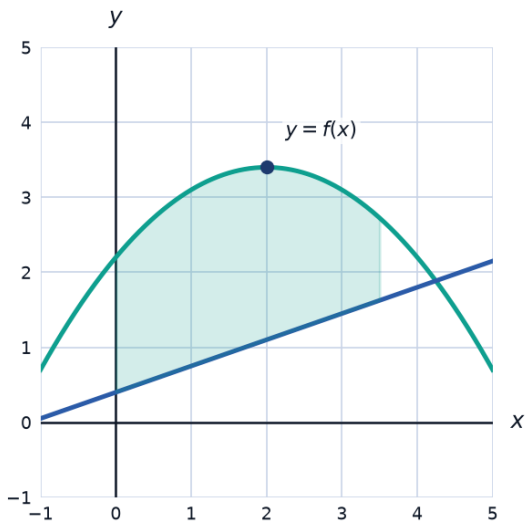
1 [Foundational] • Target: Integration & Kinematics Practice Exercises

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$. Find the area of the region enclosed between the curve and the x-axis.



2 [Foundational] • Target: Indefinite Integration Techniques

The curve C has equation $y = x^3 - 3x + 2$. The tangent to C at the point $P(2, 4)$ intersects the curve again at point Q. Find the area of the finite region completely enclosed between the curve C and the tangent line.



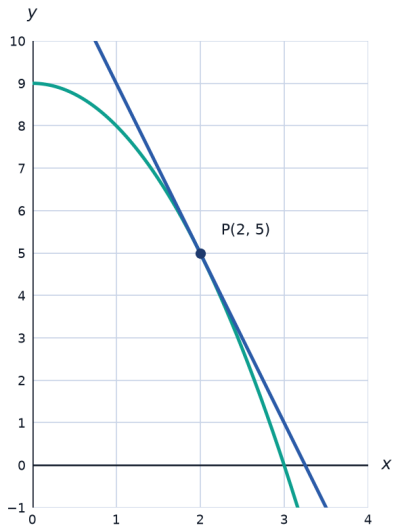
© ClipSAT — clipsat.org | All Rights Reserved

3 [Foundational] • Target: Definite Integrals and Area Under Curves

A curve has equation $y = 6x - x^2$. The straight line $y = mx$, where $0 < m < 6$, intersects the curve at the origin and at the point P. Given that the area of the region enclosed between the curve and the line is $\frac{32}{3}$, find the value of the constant m.

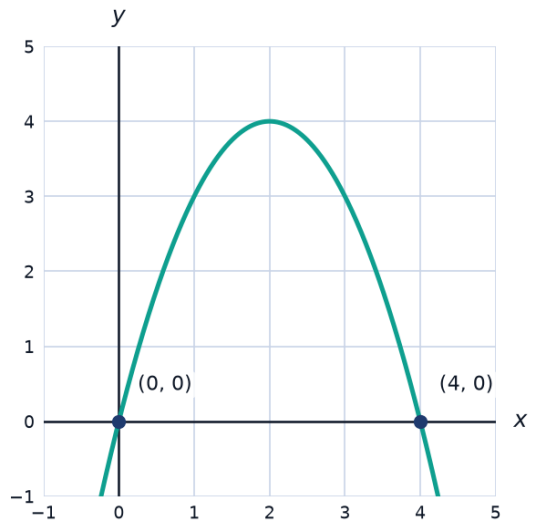
4 [Foundational] • Target: Area Between Two Intersecting Curves

The diagram shows part of the curve $y = 9 - x^2$ for $x \geq 0$ and the tangent to the curve at the point $P(2, 5)$. Find the exact area of the finite region bounded by the curve, the tangent at P, and the x-axis.



5 [Foundational] • Target: Rectilinear Motion and Kinematics

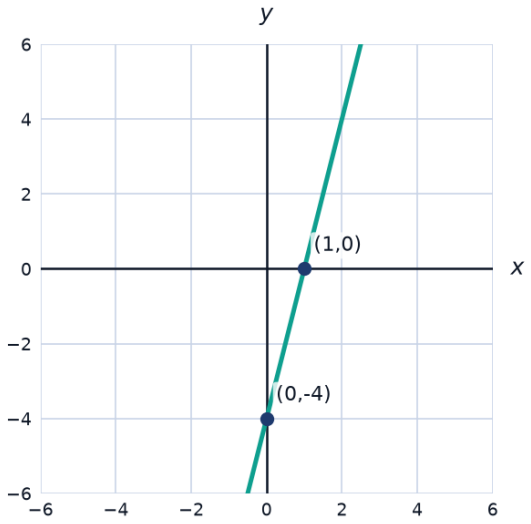
The diagram shows the curve $y = 4x - x^2$. Find the area of the region enclosed by the curve and the x-axis.



© ClipSAT — clipsat.org | All Rights Reserved

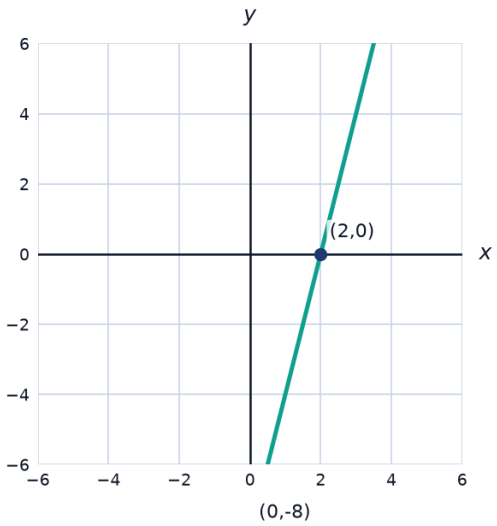
6 [Foundational] • Target: Integration & Kinematics Practice Exercises

Find the area enclosed between the curve $y = 4x - 4$, the x-axis, and the lines $x = 1$ and $x = 4$.



7 [Foundational] • Target: Indefinite Integration Techniques

Find the area enclosed between the curve $y = 4x - 8$, the x-axis, and the lines $x = 2$ and $x = 4$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Definite Integrals and Area Under Curves

Two circles, C_1 with centre O_1 and radius 6 cm, and C_2 with centre O_2 and radius $6\sqrt{3}$ cm, intersect at the points A and B. The distance between their centres is $O_1O_2 = 12$ cm. Find the exact area of the region common to both circles, in cm^2.

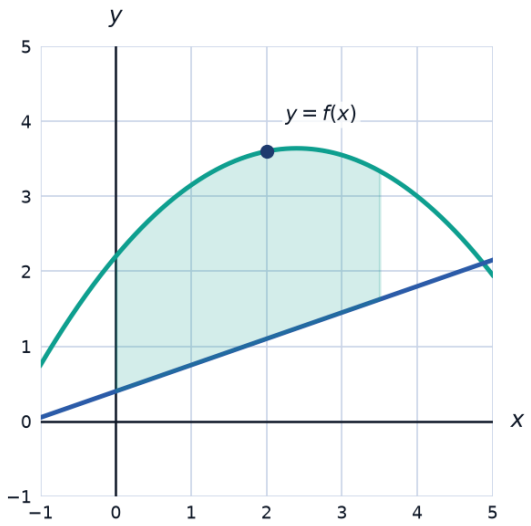
© ClipSAT — clipsat.org | All Rights Reserved

9 [Intermediate] • Target: Area Between Two Intersecting Curves

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

10 [Intermediate] • Target: Rectilinear Motion and Kinematics

Evaluate the definite integral: $\int_0^1 (9x+1)^3\,dx$.



11 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: $\int_0^1 (5x+1)^3\,dx$.

12 [Intermediate] • Target: Indefinite Integration Techniques

Evaluate the definite integral: $\int_0^1 (6x+1)^3\,dx$.

13 [Intermediate] • Target: Definite Integrals and Area Under Curves

Evaluate the definite integral: $\int_0^1 (7x+1)^3\,dx$.

14 [Intermediate] • Target: Area Between Two Intersecting Curves

Evaluate the definite integral: $\int_0^1 (8x+1)^3\,dx$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

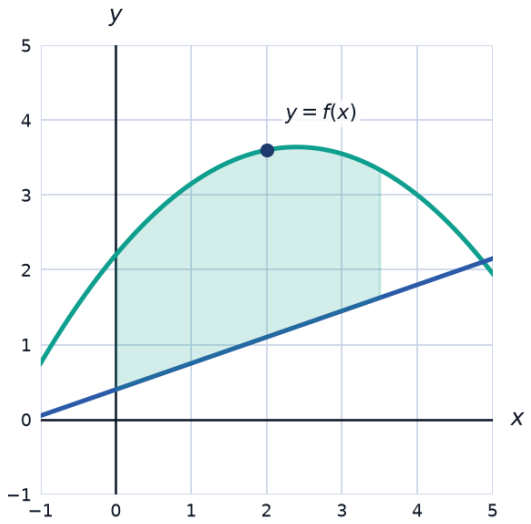
15 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Evaluate the definite integral: $\int_0^1 (3x+1)^3\,dx$.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Evaluate the definite integral: $\int_0^1 (4x+1)^3\,dx$.



17 [Challenging] • Target: Indefinite Integration Techniques • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Evaluate the definite integral: $\int_0^1 (2x+1)^3\,dx$.

18 [Challenging] • Target: Definite Integrals and Area Under Curves • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Find the area enclosed between the curve $y = 2x^2$, the x-axis, and the lines $x = 0$ and $x = 4$.

19 [Challenging] • Target: Area Between Two Intersecting Curves • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Find the area enclosed between the curve $y = x^2$, the x-axis, and the lines $x = 0$ and $x = 4$.

20 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 6.01 Permutations and Combinations

Find the area enclosed between the curve $y = 2x^2$, the x-axis, and the lines $x = 0$ and $x = 3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (2x + 1)^3 \, dx \).

a. \(\frac{80}{2} \)

b. \(\frac{84}{8} \)

c. \(\frac{80}{4} \)

d. \(\frac{80}{8} \)

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (3x + 1)^3 \, dx \).

a. \(\frac{255}{4} \)

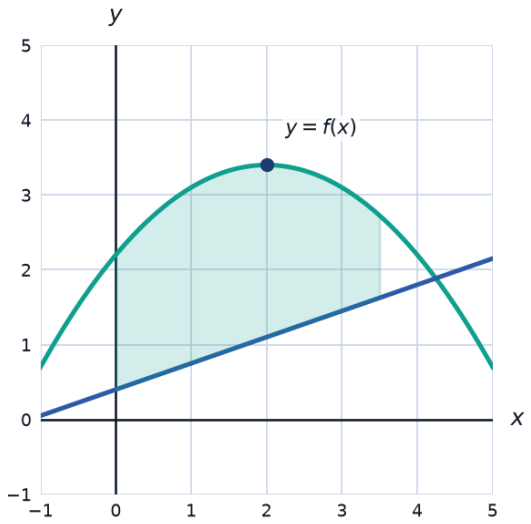
b. \(\frac{255}{3} \)

c. \(\frac{259}{12} \)

d. \(\frac{255}{12} \)

23 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (4x + 1)^3 \, dx \).



a. \(\frac{625}{16} \)

b. \(\frac{624}{4} \)

c. \(\frac{628}{16} \)

d. \(\frac{624}{16} \)

24 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (5x + 1)^3 \, dx \).

a. \(\frac{1295}{5} \)

b. \(\frac{1295}{4} \)

c. \(\frac{1295}{20} \)

d. \(\frac{1299}{20} \)

25 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (6x + 1)^3 \, dx \).

a. \(\frac{2400}{6} \)

b. \(\frac{2400}{24} \)

c. \(\frac{2404}{24} \)

d. \(\frac{2400}{4} \)

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (7x + 1)^3 \, dx \).

a. \(\frac{4095}{4} \)

b. \(\frac{4099}{28} \)

c. \(\frac{4095}{28} \)

d. \(\frac{4095}{7} \)

27 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (8x + 1)^3 \, dx \).

a. \(\frac{6560}{4} \)

b. \(\frac{6560}{8} \)

c. \(\frac{6564}{32} \)

d. \(\frac{6560}{32} \)

28 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (9x + 1)^3 \, dx \).

a. \(\frac{9999}{4} \)

b. \(\frac{9999}{9} \)

c. \(\frac{10003}{36} \)

d. \(\frac{9999}{36} \)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Indefinite Integration Techniques

Find the indefinite integral $\int (3x-1)^4\,dx$.

a. $\frac{(3x-1)^5}{15}+C$

b. $\frac{(3x-1)^5}{5}+C$

c. $\frac{(3x-1)^3}{3}+C$

d. $12(3x-1)^3+C$

30 [Intermediate] • Target: Indefinite Integration Techniques

Evaluate $\int (e^{2x}+\sin 3x)\,dx$.

a. $\frac{1}{2}e^{2x}-\frac{1}{3}\cos 3x+C$

b. $\frac{1}{2}e^{2x}+\frac{1}{3}\cos 3x+C$

c. $2e^{2x}-3\cos 3x+C$

d. $e^{2x}-\cos 3x+C$

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Indefinite Integration Techniques

A curve passes through $(1, 4)$ and satisfies $\frac{dy}{dx} = 4x - 1$. Find the equation of the curve.

a. $y = 2x^2 - x + 3$

b. $y = 2x^2 - x + 4$

c. $y = 4x^2 - x + 1$

d. $y = 2x^2 + x + 1$

32 [Intermediate] • Target: Definite Integrals and Area Under Curves

Evaluate the exact value of the definite integral $\int_1^4 \left(2x + \frac{1}{\sqrt{x}}\right) dx$.

a. 17

b. 15

c. 19

d. 16

33 [Intermediate] • Target: Definite Integrals and Area Under Curves

Find the area enclosed by the curve $y = 6x - x^2$ and the x-axis.

a. 36

b. 18

c. 72

d. 24

34 [Intermediate] • Target: Definite Integrals and Area Under Curves

Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$.

a. 1

b. $\sqrt{2}$

c. $\frac{1}{2}$

d. $\pi/4$

35 [Intermediate] • Target: Area Between Two Intersecting Curves

Find the area of the finite region enclosed between the parabola $y = x^2$ and the line $y = 2x$.

a. $\frac{4}{3}$

b. $\frac{2}{3}$

c. 2

d. $\frac{8}{3}$

36 [Intermediate] • Target: Area Between Two Intersecting Curves

Find the area between the curves $y = 4 - x^2$ and $y = x^2 - 4$.

a. $\frac{64}{3}$

b. $\frac{32}{3}$

c. 16

d. 8

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Area Between Two Intersecting Curves

Set up and evaluate the integral representing the area between $y = \sqrt{x}$ and $y = x$ for $0 \le x \le 1$.

a. $\frac{1}{6}$

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{2}{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A particle moves in a straight line with velocity $v(t) = 3t^2 - 12t + 9$ m/s. Find the displacement from $t = 0$ to $t = 3$.

a. 0 m

b. 9 m

c. 18 m

d. 6 m

39 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Find the acceleration of the particle in question 10 at $t = 4$ seconds.

a. 12 m/s^2

b. 6 m/s^2

c. 24 m/s^2

d. 0 m/s^2

40 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 6.01 Permutations and Combinations

Calculate the total distance travelled by the particle with $v(t) = 3(t - 1)(t - 3)$ from $t = 0$ to $t = 3$.

a. 8 m

b. 0 m

c. 4 m

d. 12 m

41 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Find the area enclosed between the curve $y = 3x$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 37/2

b. 27/2

c. 27

d. 25/2

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Indefinite Integration Techniques • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Find the area enclosed between the curve $y = 3x$, the x-axis, and the lines $x = 0$ and $x = 5$.

a. 85/2

b. 75/2

c. 73/2

d. 75

43 [Challenging] • Target: Definite Integrals and Area Under Curves • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the area enclosed between the curve $y = 3x^2$, the x-axis, and the lines $x = 0$ and $x = 5$.

a. 124

b. 130

c. 250

d. 125

44 [Challenging] • Target: Area Between Two Intersecting Curves • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Find the area enclosed between the curve $y = 3x^2$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 54

b. 32

c. 26

d. 27

45 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Find the area enclosed between the curve $y = x$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 19/2

b. 9

c. 9/2

d. 7/2

© ClipSAT — clipsat.org | All Rights Reserved